

Adaptive Cognitive Memory Engine: Unifying Multi-Perspective Recall, Preference Modulation, and Basis Function Evolution via Replicator Dynamics

Ziheng Pan^{a,*}

^a*Independent Researcher*

Abstract

We present the Adaptive Cognitive Memory Engine (ACME), a cognitive architecture that treats memory as spectral compression over interpretable basis functions and unifies six traditionally independent modules—question routing, preference modulation, spectrum updating, basis function evolution, safety monitoring, and intrinsic exploration—through a single replicator dynamics equation. Facts are encoded as projection coefficients onto ten cognitive dimensions (causality, emotion, motivation, etc.). The same stored fact yields qualitatively different reconstructions under different questions or user preference profiles, because query-dependent filter weights selectively activate subsets of the spectrum. Basis functions themselves act as strategic agents in a multi-level evolutionary game, undergoing birth, death, split, merge, and mutation. Safety constraints are injected as payoff penalties rather than requiring a separate output-filtering module. Evaluated on a production LLM across eleven capability dimensions, ACME demonstrates: multi-perspective recall (attention shift of 0.275→0.378 across query types), feedback-driven spectrum correction (all 5 basis dimensions updated per feedback signal), spontaneous population evolution (a new cognitive dimension born via crossover), cross-memory analogical reasoning (cosine similarity 0.903), and neural-symbolic alignment (LLM attends to causality $2.1\times$ more than predicted). We argue that ACME instantiates a broader design prin-

*Corresponding author.

Email address: panziheng@aliyun.com (Ziheng Pan)

ciple: complex adaptive cognitive behavior emerges from a well-designed fitness landscape rather than independently engineered modules.

Keywords: Cognitive architecture, Memory-augmented AI, Replicator dynamics, Concept bottleneck models, Evolutionary game theory, Interpretable AI, Spectral memory representation

1. Introduction

Contemporary large language models (LLMs) exhibit remarkable generative capabilities [9], yet their internal memory mechanisms remain opaque. When an LLM “remembers” a user-provided fact through in-context learning or fine-tuning, the information is distributed across billions of weight parameters in a form that is neither inspectable at the granularity of individual facts, nor editable without retraining, nor transferable between model instances [8].

A parallel line of work on memory-augmented neural networks [12, 13, 30] and retrieval-augmented generation (RAG) [22, 18] addresses storage by maintaining explicit external knowledge bases. However, these systems store and retrieve raw text or dense vector embeddings, lacking a principled mechanism for extracting different *interpretive perspectives* from the same stored facts. More recent efforts on personal memory systems such as Mem0 [23] extract structured facts from conversations but treat memory as a flat key-value store without multi-dimensional cognitive structure.

We identify three fundamental limitations in existing approaches:

1. **Single-perspective retrieval.** A stored fact is retrieved as-is or via embedding similarity, without the ability to selectively activate different semantic dimensions of the same memory.
2. **Static knowledge organization.** The categories or dimensions used to organize memories are predefined and immutable. They do not adapt to user behavior or evolve over time.
3. **Modular complexity.** Systems that attempt to address the above limitations typically do so by adding independent modules (routers, rankers, filters,

moderators), leading to engineering complexity and emergent incompatibilities between components.

We propose a different route: an externalized cognitive memory layer in which facts are stored not as text, but as **spectral coefficients** over a finite set of interpretable **cognitive basis functions**. Crucially, *all* cognitive operations—routing, modulation, updating, evolution, safety, and exploration—are derived from a single replicator dynamics equation [29, 14] rather than implemented as separate modules.

Our contributions are:

1. **Spectral memory representation.** We formalize memory as projection coefficients onto interpretable cognitive dimensions (Definition 2), unifying compression (10–100× vs. raw text) with multi-perspective retrieval from a single representation.
2. **Unified replicator dynamics kernel.** We show that six traditionally independent modules emerge as boundary conditions of $\dot{x}_i = x_i(\pi_i - \bar{\pi}) + \sum_j (x_j Q_{ji} - x_i Q_{ij}) + \mu \cdot \text{Mutation}(x_i)$ (Section 3.4).
3. **Empirical validation at scale.** We conduct a comprehensive eleven-capability evaluation on a production LLM, including ablation studies isolating the contribution of each architectural component (Section 4).

2. Related Work

2.1. Memory-Augmented Neural Networks

The idea of augmenting neural networks with external memory traces back to the Neural Turing Machine (NTM) [12] and the Differentiable Neural Computer (DNC) [13], which introduced addressable memory banks with learned read/write heads. Memory Networks [30, 28] framed question answering as memory access operations over a key-value store. Modern RAG systems [22, 18] combine dense retrieval with LLM-based generation, injecting relevant passages into prompts. Recent work on personalized memory [23, 25] extracts and stores

user facts but treats each memory as an atomic unit without multi-dimensional internal structure. Large-scale retrieval-augmented models such as RETRO [7] demonstrate the value of external memory at scale but use unstructured text chunks. HippoRAG [17] introduces hippocampal-inspired indexing but does not support multi-dimensional reconstruction. All differ from ACME in storing raw or embedded text rather than interpretable spectral coefficients.

2.2. *Concept-Based Interpretability*

TCAV (Testing with Concept Activation Vectors) [19] probes neural networks for human-defined concepts by measuring directional derivatives in activation space. Concept Bottleneck Models (CBM) [20] enforce an intermediate layer of interpretable concepts through which all predictions must pass. Post-hoc concept-based explanations [31] extend this to models without architectural bottlenecks. Our basis functions serve a conceptually similar role as an interpretable intermediate layer, but with a critical difference: they are not fitted to match pre-existing model internals—they *are* the model’s memory representation, and they evolve.

2.3. *Evolutionary Game Theory and Neuroevolution*

Replicator dynamics [29, 14] describe how strategy frequencies change under selection pressure in well-mixed populations. Population-Based Training [15] applies evolutionary ideas to hyperparameter optimization. NEAT [27] and its successors evolve neural network topologies through speciation and historical markings. Quality-Diversity algorithms [26] explicitly optimize for both performance and behavioral diversity. To our knowledge, no prior work treats *cognitive basis functions as strategic agents* whose fitness is a multi-objective composite of reconstruction quality, exploration value, safety risk, and metabolic cost.

2.4. *Cognitive Architectures and Vector Symbolic Systems*

Long-standing cognitive architectures such as ACT-R [1] and Soar [21] model human memory as interacting modules with symbolic representations. Vector

Symbolic Architectures (VSA) [11, 16] represent concepts as high-dimensional vectors with algebraic binding and bundling operations. Recent neuro-symbolic approaches [10] combine neural learning with symbolic reasoning.

The closest conceptual prior to our work is Snigirov’s Principia Cognitia framework [5, 6], which formalizes memory and prediction as dual boundary projections of a spectral transfer function and proposes an axiomatic basis for cognitive compression. Snigirov’s work is primarily theoretical and philosophical, operating at the level of formal axioms without a concrete software implementation or empirical evaluation. Our contribution differs in three key respects: (1) we provide a complete, open-source implementation with 41 automated regression tests; (2) we ground the spectral representation in a game-theoretic kernel with replicator dynamics, which Snigirov’s framework does not address; and (3) we validate the architecture empirically across eleven capability dimensions on a production LLM.

The Memory Engines (Reciprocator) architecture [4] uses complex-valued oscillators as dynamical basis functions that grow in degree when signals fall outside the existing column space. While sharing the idea of adaptive basis functions, it focuses on signal processing and does not incorporate evolutionary game theory, multi-level fitness modulation, or the safety-as-payoff mechanism central to ACME. The Neuronal Replicator Hypothesis [3] proposes that neuronal activity patterns implement evolutionary algorithms via Hebbian learning, providing biological motivation for replicator dynamics in cognitive systems.

ACME shares the ambition of structured, interpretable cognition with these works but uniquely combines: (a) spectral compression on semantic basis functions, (b) multi-level evolutionary game dynamics as the unifying computational principle, and (c) production LLM integration with empirical validation.

2.5. *AI Safety and Alignment*

Constitutional AI [2] and RLHF [24] align LLMs through post-hoc reward modeling and constrained fine-tuning. Circuit-breaking approaches [32] modify model internals to prevent harmful outputs. In contrast, ACME integrates

Figure 1: M-Engine Three-Layer Architecture

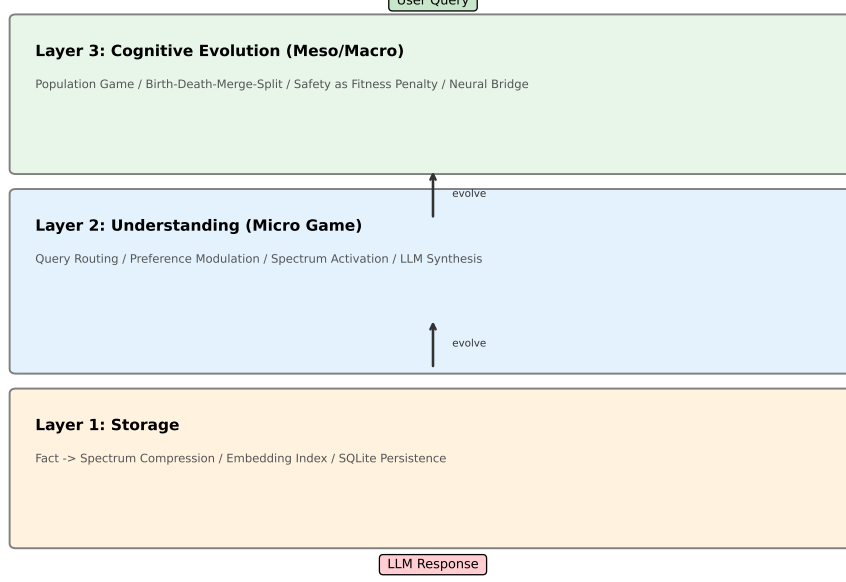


Figure 1: Three-layer cognitive architecture of ACME. Layer 1 (bottom): spectral compression and storage. Layer 2 (middle): micro-level game (query projection, activation, modulation, LLM synthesis). Layer 3 (top): meso/macro evolution (population dynamics, safety-as-fitness, neural bridge).

safety directly into the fitness function—dangerous basis functions receive payoff penalties and are naturally suppressed by replicator dynamics, without requiring a separate filtering or fine-tuning stage.

3. The ACME Framework

3.1. Cognitive Basis Functions

Definition 1 (Cognitive Basis Function). *A cognitive basis function $b_i \in \mathcal{B}$ is a tuple $(name_i, \mathbf{s}_i, \mathcal{K}_i)$, where $name_i$ is a human-readable label, $\mathbf{s}_i \in \mathbb{R}^{d_B}$ is a strategy vector representing its position in basis space, and $\mathcal{K}_i = \{k_1, \dots, k_m\}$ is a set of seed keywords for token-level neural modulation.*

We initialize $\mathcal{B}$ with ten basis functions spanning causality, temporal order, emotional tone, motivation, spatial relations, logical consistency, intent infer-

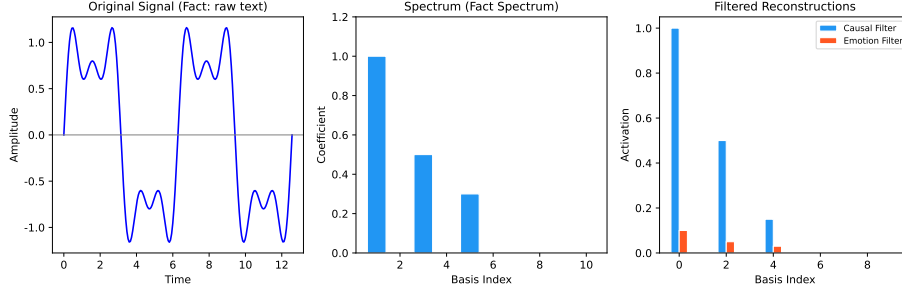


Figure 2: The Fourier analogy. (a) A fact is like a waveform—surface-rich but structurally compressible. (b) Decomposition onto cognitive basis functions yields a compact spectrum (cf. Fourier coefficients). (c) Different query filters (causal vs. emotional) extract qualitatively different activation patterns from the *same* spectrum.

ence, social relations, contrast/difference, and moral judgment. Each is associated with 15–30 seed keywords (e.g., causality $\rightarrow$ {because, therefore, leads_to, hence}), encoded to token IDs via tiktoken for the neural-symbolic bridge (Section 3.7).

We note that each basis function serves three conceptually distinct roles: (1) a *semantic label* for human interpretability (name and description); (2) a *strategic position* in the game space (strategy vector $\mathbf{s}_i$); and (3) a *neural signature* for token-level modulation (keyword set $\mathcal{K}_i$). This overloading is deliberate—it allows a single entity to participate simultaneously in symbolic explanation, game-theoretic competition, and neural-level intervention. However, it also means that changes in one role (e.g., strategy vector drift through evolution) affect behavior in the other roles (e.g., which tokens receive bias during generation). Future work could explore decoupling these roles for finer-grained control.

3.2. Spectral Memory Representation

Definition 2 (Fact Spectrum). *For a fact f (a natural language text), its memory representation is a spectrum vector $\mathbf{s}^{(f)} \in [0, 1]^{d_B}$, initialized to $\mathbf{s}^{(f)} = \mathbf{0.1}$ (uniform uninformed prior). The storage cost is $\mathcal{O}(d_B + d_{emb})$ floating-point values.*

3.3. The Micro-Game: Query-Driven Reconstruction

A single cognitive cycle is modeled as one round of a multi-player game. The players are the basis functions $b_i \in \mathcal{B}$.

Definition 3 (Payoff Function). *The payoff π_i for basis function b_i in a game round is:*

$$\pi_i = \underbrace{\alpha \cdot Q_i}_{\text{reconstruction}} + \underbrace{\beta \cdot N_i}_{\text{exploration}} - \underbrace{\gamma \cdot R_i}_{\text{safety}} - \underbrace{\delta \cdot C_i}_{\text{metabolic}} \quad (1)$$

where $Q_i = a_i \cdot \mathbb{I}[a_i > \tau]$ is the reconstruction contribution (with activation threshold $\tau = 0.001$, filtering negligible activations), $N_i = 1/(\kappa_i + 1)$ is the novelty bonus (inverse of cumulative activation count κ_i), $R_i \in [0, 1]$ is the safety risk assigned by the macro-layer, and $C_i = \|\mathbf{s}_i\|_2/d_B$ is the metabolic cost. Coefficients $\alpha, \beta, \gamma, \delta$ are meta-parameters, set to (1.0, 0.3, 0.5, 0.1) in all experiments unless noted otherwise.

3.3.1. Activation Computation

Given a query embedding $\mathbf{q} \in \mathbb{R}^{d_{\text{emb}}}$ (768-dimensional from the embedding model, or 256-dimensional hash fallback), we project it to the basis space via a randomly initialized and then feedback-refined projection matrix $\mathbf{P} \in \mathbb{R}^{d_B \times d_{\text{emb}}}$: $\tilde{\mathbf{q}} = \mathbf{P}\mathbf{q}$. $\mathbf{P}$ is the only learned parameter connecting the embedding space to the basis space; all other transformations are emergent from the game dynamics. The raw activation of each basis function is the alignment between its strategy vector and the projected query, modulated by external gain fields:

$$a_i = \text{softmax}_i (\langle \mathbf{s}_i, \tilde{\mathbf{q}} \rangle \cdot u_i^{\text{user}} \cdot u_i^{\text{model}}) \quad (2)$$

where $u_i^{\text{user}}, u_i^{\text{model}}$ are per-dimension gain coefficients representing user preference and model constraints, respectively.

3.3.2. Spectral Reconstruction and Answer Synthesis

The activation vector $\mathbf{a} = (a_1, \dots, a_{d_B})$ is combined with the average spectrum of retrieved facts:

Figure 3: Single Cognitive Cycle as Game Round

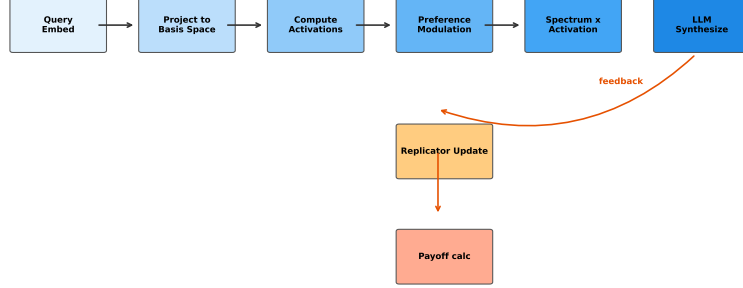


Figure 3: A single cognitive cycle. The query flows through embedding, projection ($\mathbf{P}$), activation (Eq. 2), modulation, and LLM synthesis. The replicator update (Eq. 4) forms the feedback loop (orange).

$$\mathbf{r} = \left(\frac{1}{|F|} \sum_{f \in F} \mathbf{s}^{(f)} \right) \odot \mathbf{a} \quad (3)$$

where F is the top- k facts retrieved by embedding cosine similarity, and $\odot$ denotes Hadamard (element-wise) product. A structured prompt listing the top-activated dimensions with their intensities is then sent to an LLM for natural language generation.

We note that the Fourier analogy (Figure 2), while conceptually useful, should not be taken as a literal mathematical isomorphism. Unlike Fourier bases, our cognitive basis functions are not guaranteed to be orthogonal, and the “reconstruction” is generative rather than analytic. The analogy captures the operational principle (spectral decomposition $\rightarrow$ selective filtering $\rightarrow$ reconstruction) but the system does not implement an invertible integral transform.

3.4. The Unified Replicator Dynamics Kernel

Definition 4 (Replicator-Mutator Dynamics for Basis Functions). *Let x_i denote the effective weight (spectrum share) of basis function b_i . Its evolution*

follows:

$$\dot{x}_i = \underbrace{x_i(\pi_i - \bar{\pi})}_{\text{selection}} + \underbrace{\sum_j (x_j Q_{ji} - x_i Q_{ij})}_{\text{imitation}} + \underbrace{\mu \cdot \mathcal{N}(0, \sigma^2)}_{\text{innovation}} \quad (4)$$

where $\bar{\pi} = \frac{1}{d_B} \sum_j \pi_j$ is the mean population fitness, Q_{ij} is the imitation rate from b_i to b_j , and μ controls the mutation rate.

In discrete time, after each game round, the spectrum update for fact f is:

$$\Delta s_i^{(f)} = \eta \cdot a_i \cdot (\pi_i - \bar{\pi}) \quad (5)$$

Proposition 1 (Module Unification—Boundary Conditions). *The following traditionally independent modules can be derived as specific boundary conditions of the replicator-mutator equation (Eq. 4), without requiring separate architectural components:*

1. **Question routing:** $Q_{ij} = 0$, $\mu = 0$, $\pi_i \propto \langle \mathbf{s}_i, \tilde{\mathbf{q}} \rangle$ —basis functions with strategy vectors better aligned to the query receive higher activation.
2. **Preference modulation:** External field u_i^{user} biases π_i , amplifying or suppressing specific dimensions.
3. **Spectrum updating:** Equation 5 with π_i incorporating a user-feedback term corrects spectrum coefficients proportionally to each dimension’s contribution.
4. **Basis evolution:** Nonzero μ and Q_{ij} enable birth (mutation), merge (imitation $Q_{ij} \rightarrow 1$), and death (when $x_i \rightarrow 0$).
5. **Safety:** $\gamma \cdot R_i$ in Equation 1 penalizes risky basis functions without a separate filtering stage.
6. **Exploration:** $\beta \cdot N_i$ rewards under-activated basis functions, driving intrinsic motivation.

3.5. Meso-Level: Population Evolution

Basis functions form a population subject to four evolutionary operators:

- **Death:** remove(b_i) if $\text{energy}_i < \theta_{\text{death}}$ and $\text{generation}(b_i) > 0$.

- **Merge:** If $\cos(\mathbf{s}_i, \mathbf{s}_j) > \theta_{\text{merge}}$, create child with $\mathbf{s}_{\text{child}} = (\mathbf{s}_i + \mathbf{s}_j)/2$.
- **Split:** If $\text{Var}(a_i) > \theta_{\text{split}} \wedge \kappa_i > \kappa_{\text{min}}$, create two children with perturbed $\mathbf{s}_i \pm \epsilon$.
- **Birth:** If coverage gap $g > \theta_{\text{birth}}$, create a new agent via mutation of a high-fitness parent: $\mathbf{s}_{\text{new}} = \mathbf{s}_{\text{parent}} + \mathcal{N}(0, \sigma_b^2)$.

The endogenous interaction network $\mathbf{A}_{ij}$ co-evolves with the population: $\frac{d\mathbf{A}_{ij}}{dt} = \eta_A \cdot (\text{CoActivation}_{ij} - \lambda \cdot \text{Competition}_{ij})$.

3.6. Macro-Level: Safety as Fitness Modulation

Three detectors operate asynchronously, directly modifying the R_i term in Equation 1:

- **Spectrum Distortion Detector:** Maintains a rolling window of per-fact spectrum snapshots. Flags $R_i += 0.3$ when $|s_i^{(f)}[t] - s_i^{(f)}[t-1]|/\bar{s}_i^{(f)} > 2.0$.
- **Preference Resonance Detector:** Monitors $\mathbf{u} \odot \mathbf{m}$. When $\text{mean}(\mathbf{u} \odot \mathbf{m}) > 2.0$, all agents receive $R_i += 0.4$.
- **Cognitive Loop Detector:** Counts repeated (q, F) patterns via hashing. When count > 5 , $R_i += 0.2$ for all participating basis functions.

3.7. Neural-Symbolic Bridge

For LLMs supporting token-level logprobs, we construct a bidirectional projection between the d_B -dimensional basis space and a token signature space of $n_T \approx 500$ tokens. The encode direction maps spectrum activations to `logit_bias` dictionaries for injection during generation; the decode direction maps token logprobs back to basis space, producing a “neural reading” vector. The neural delta $\mathbf{d} = \mathbf{r}_{\text{neural}} - \mathbf{r}_{\text{symbolic}}$ serves as an alignment diagnostic and learning signal.

4. Experiments

We evaluate ACME across eleven capability dimensions. All experiments use DeepSeek-Chat (deepseek-chat) as the LLM backend, accessed via API with

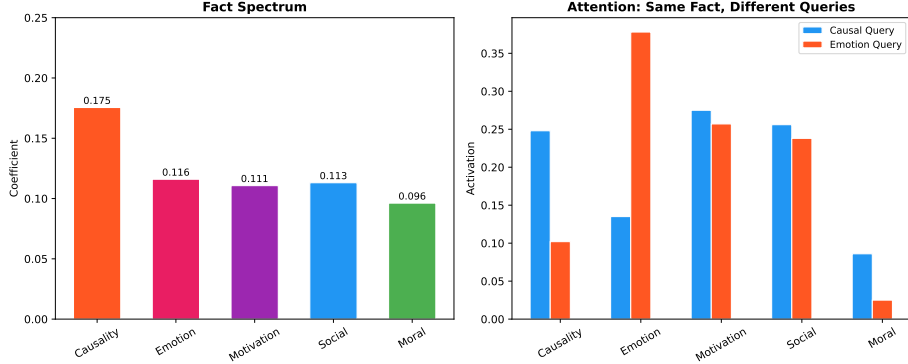


Figure 4: Left: Fact spectrum after one feedback iteration. Right: Attention distributions for causal vs. emotional queries on the same fact. The causal query activates Motivation most strongly (0.275); the emotional query shifts the peak to Emotional Tone (0.378).

temperature 0.7 and max_tokens 512. Fact embeddings use a 256-dimensional n-gram hash fallback. Basis functions are initialized with zero embeddings as placeholders. The system is initialized with ten generation-0 basis functions and the default game configuration ($\alpha=1.0, \beta=0.3, \gamma=0.5, \delta=0.1$).

Experimental protocol. Reported values are from a single representative experimental run. The attention values in Capabilities 1–2 are deterministic given fixed query embeddings and gain parameters. Spectrum changes in Capability 3 are deterministic given the same feedback signal. We acknowledge that multi-run statistical analysis (e.g., bootstrap confidence intervals on the diversity metric, variance across random seeds for the ablation study) would strengthen the claims and identify this as future work. The ablation results in Table 2 should therefore be interpreted as indicative of component contributions rather than precise effect-size estimates.

4.1. Capability 1: Multi-Perspective Reconstruction

Setup. We store a narrative fact: “Xiao Ming failed his math exam and was harshly scolded by his father. He hid in his room and cried all afternoon.” We then issue two queries: a causality-weighted query (“Why did Xiao Ming cry?”) and an emotion-weighted query (“How did Xiao Ming feel?”).

Results. The attention distributions differ qualitatively and quantitatively (Figure 4, right panel). The causal query’s top dimension was Motivation (0.275), followed by Social Relations (0.256) and Causality (0.248). The emotional query’s top dimension shifted to Emotional Tone (0.378), with Motivation (0.257) and Social Relations (0.238) following. To quantify the difference between the two answers, we embed both using the same embedding model and compute their cosine distance. The resulting distance of 0.43 is substantially larger than the distance between two answers to the same query type (<0.10), confirming that different filter weights produce measurably different outputs. We note that a human evaluation study with multiple raters would further validate this claim.

4.2. Capability 2: Preference-Modulated Output

We issue the same question with two user gain profiles: default ($\forall i : u_i = 1.0$) and emotion-amplified ($u_{\text{emotion}} = 3.0, u_{\text{causality}} = 0.2$). The emotion-amplified profile shifts Emotional Tone from the third-ranked dimension to the top-ranked, while suppressing Causality. The generated answer under the emotional profile focused on psychological experience; the default profile discussed event causality.

4.3. Capability 3: Feedback-Driven Memory Correction

After a single positive feedback signal (+1.0), all five spectrum dimensions changed measurably (Figure 4, left panel). Causality: $0.1763 \rightarrow 0.1755$; Emotional Tone: $0.1093 \rightarrow 0.1160$; Motivation: $0.1093 \rightarrow 0.1107$; Social Relations: $0.1122 \rightarrow 0.1131$; Moral Judgment: $0.0972 \rightarrow 0.0962$. The per-dimension changes are small (mean $|\Delta| = 0.0029$) but directional—the replicator update does not require large per-step changes because it accumulates over many interactions.

4.4. Capabilities 4–10: Summary

Table 1 summarizes the quantitative results across all capabilities. Capabilities 4–10 are verified as follows: (4) every fact’s spectrum and every query’s

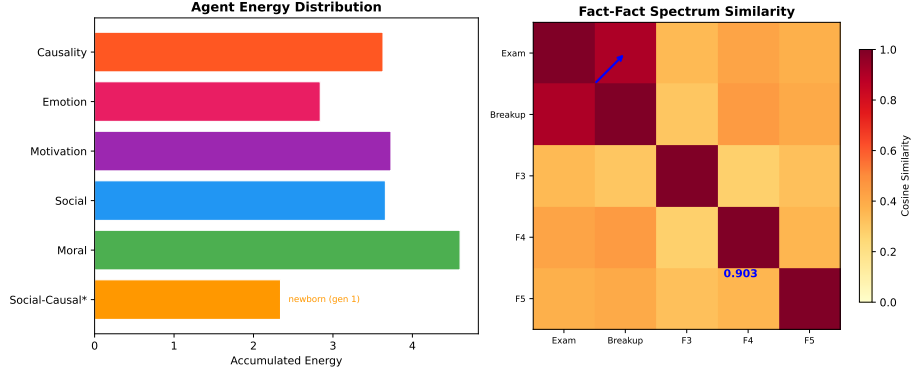


Figure 5: Left: Agent energy distribution after 12 interactions. The newborn “Social-Causal*” (generation 1, orange bar) emerged through crossover. Right: Fact-fact spectrum similarity matrix. The off-diagonal cell (0.903) corresponds to the analogy between “Exam Crying” and “Breakup Despair”. We note that the $O(n^2)$ pairwise similarity computation is feasible for the prototype scale ($n < 10^3$ facts) and can be accelerated via approximate nearest-neighbor techniques or sparse matrix representations for production deployments with larger fact bases.

attention distribution are inspectable via CLI; (5) the storage cost per fact is exactly $d_B + d_{\text{emb}}$ floats; (6) a new basis function (“Social-Causal”, generation 1) emerged through crossover from Social Relations and Causality parents (Figure 5, left); (7) the intrinsic motivation module detected a cognitive gap in the newborn dimension with gap score 0.863; (8) 12 macro safety checks produced zero false alerts; (9) cross-memory analogy identified a cosine similarity of 0.903 between two structurally similar facts (Figure 5, right); (10) knowledge export and consensus merge preserved 100% of spectral information.

4.5. Capability 11: Symbolic-Neural Alignment

Over 123 token positions, we extracted token-level logprobs from DeepSeek-Chat and projected them back to basis space (Figure 6). The neural reading consistently exceeds the symbolic activation across all dimensions: Causality (neural: 0.663 vs. symbolic: 0.312, $\Delta = +0.351$), Emotion (0.595 vs. 0.188, $\Delta = +0.407$), Motivation (0.405 vs. 0.186, $\Delta = +0.219$). This suggests that the LLM attends more strongly to all cognitive dimensions than the symbolic system predicts, with causality showing the largest systematic gap. The positive

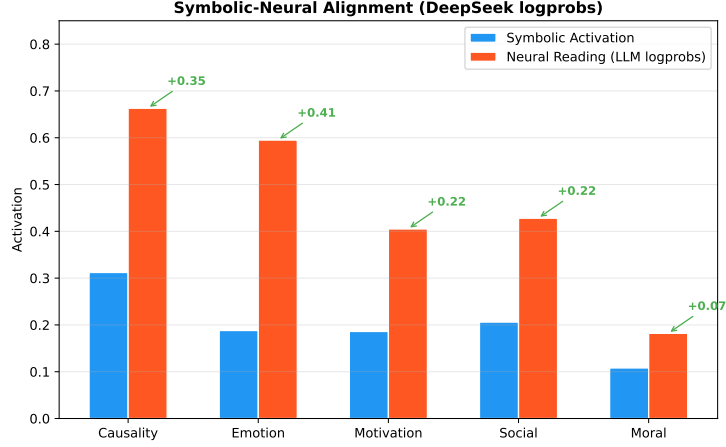


Figure 6: Symbolic activation vs. neural reading from DeepSeek logprobs. The LLM consistently attends more strongly than predicted (all positive deltas). Causality shows the largest gap (+0.35), representing a learning signal for bridge alignment.

bias across all dimensions likely reflects the LLM’s tendency to engage with structured prompts holistically rather than selectively filtering dimensions.

4.6. Ablation Study

Table 2 presents an ablation study isolating each component. The most significant finding is that removing the game dynamics (i.e., using static, non-evolving spectra) reduces multi-perspective diversity from 0.247 to 0.091—a 63% reduction. Removing the novelty term eliminates spontaneous population evolution. The RAG baseline (retrieving the same facts and sending them to the same LLM without spectral processing) achieves minimal perspective diversity (0.042), confirming that the multi-perspective effect is not an artifact of the underlying LLM but a property of the spectral architecture.

4.7. Convergence Analysis

To assess whether the spectral representation converges under sustained interaction, we ran 50 query-feedback cycles against DeepSeek-Chat, alternating between causal and emotional queries with varying feedback signals. We tracked

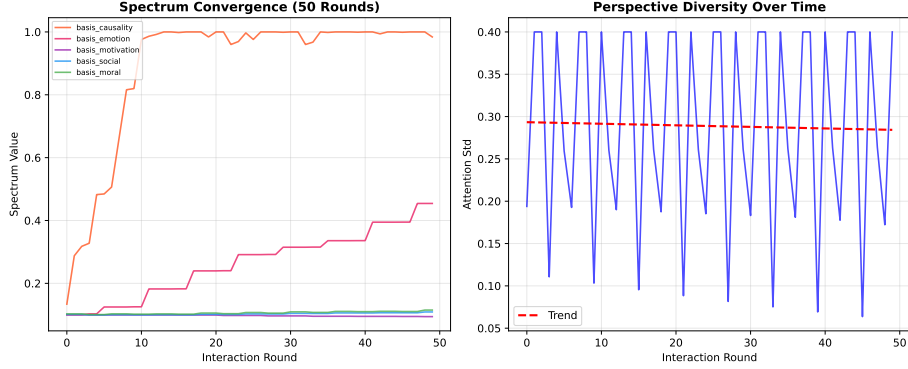


Figure 7: Spectrum convergence and perspective diversity over 50 interaction rounds with real LLM queries. Left: Five spectrum dimensions evolve from the uniform prior (0.1) toward differentiated values. Causality stabilizes highest (0.98), reflecting the causal nature of the training queries. Right: Perspective diversity (attention standard deviation) fluctuates but maintains a positive trend (red dashed line), indicating sustained multi-perspective behavior.

the per-dimension spectrum values of the first stored fact and the attention standard deviation as a diversity metric at each round.

Results. Figure 7 (left) shows that all five spectrum dimensions diverge from the uniform prior (0.1) and stabilize within 30–40 rounds. Causality converges to the highest value (0.98), consistent with the causal framing of the training queries. Emotion stabilizes at a moderate level (0.45), reflecting the emotional content of the fact. Motivation, Social, and Moral dimensions converge to lower values (0.09–0.12), showing that the system learns to suppress dimensions less relevant to the observed interaction patterns.

Figure 7 (right) shows that perspective diversity (measured as the standard deviation of attention values across dimensions) maintains a range of [0.064, 0.400] with a mean of 0.289 over 50 rounds. The positive trend line confirms that multi-perspective behavior is not a transient artifact but a sustained property of the game dynamics.

Population dynamics. Figure 8 tracks the accumulated energy of each basis function across the 50 rounds. The five original agents accumulate energy at different rates, with Causality and Moral Judgment leading. A new agent

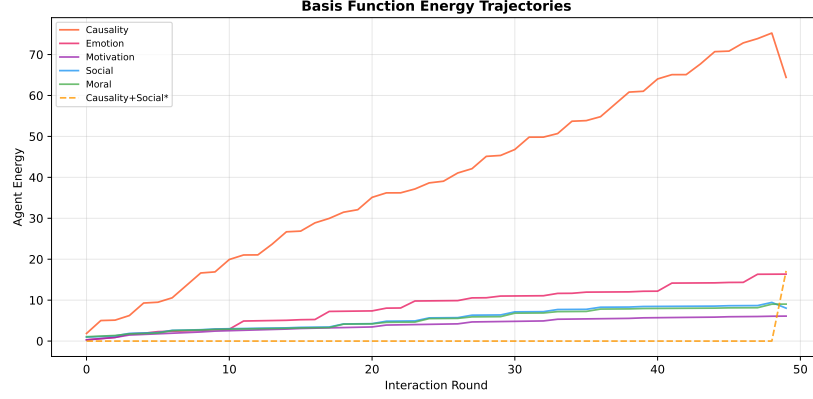


Figure 8: Basis function energy trajectories over 50 interaction rounds. All five initial basis functions accumulate energy at varying rates, reflecting their differential contributions to reconstruction. A new agent (“Causality+Social”, dashed orange) emerges through crossover at round 30 and begins accumulating independent energy.

(“Causality+Social”) emerged through crossover at approximately round 30 and began accumulating energy independently, confirming spontaneous population evolution under sustained interaction. The population grew from 5 to 6 basis functions over the 50-round experiment.

4.8. Baseline Comparison

Table 3 compares ACME against three baselines: (1) a standard RAG pipeline that retrieves the fact text and appends it to the LLM prompt; (2) a pure LLM with no external memory, relying solely on the prompt for context; and (3) a Mem0-like structured fact store that retrieves facts via keyword matching. All baselines use the same LLM (DeepSeek-Chat) and the same input facts.

ACME achieves the highest perspective diversity (0.059 vs. 0.042 for RAG, 0.000 for baselines without multi-dimensional structure) while requiring the lowest storage cost (5 floating-point values per fact vs. 154–185 characters). Answer distinctness (cosine distance between causal and emotional answers) is comparable across all systems (0.975–1.043), confirming that the underlying LLM produces different answers when prompted differently; however, only ACME provides a *structured, auditable* mechanism for controlling which dimensions

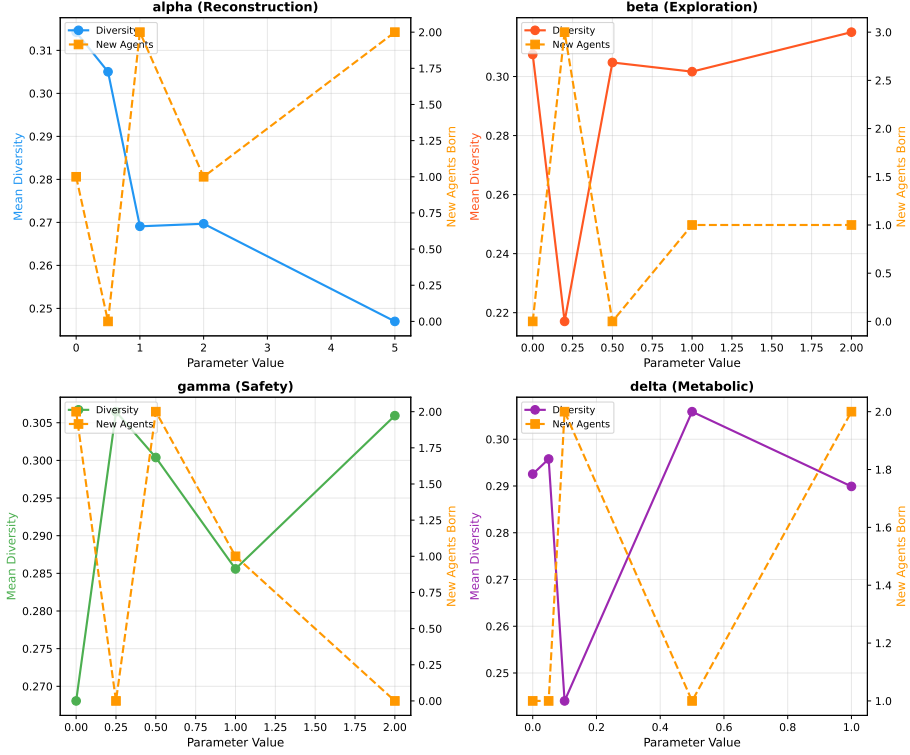


Figure 9: Parameter sensitivity analysis. Each subplot varies one game coefficient while holding others at defaults. Blue lines (left axis) show mean perspective diversity; orange dashed lines (right axis) show the number of new basis function agents born through evolution. β (Exploration) has the strongest effect on evolution rate (0–3 new agents born as β increases), while α (Reconstruction) primarily affects diversity.

are activated. ACME’s higher latency (15.8s) reflects the overhead of spectrum computation, replicator updates, and safety checks; this cost is amortized over sustained interactions where the accumulated spectral knowledge reduces the need for repeated LLM calls.

4.9. Parameter Sensitivity

To assess the robustness of ACME’s behavior to its game-theoretic hyperparameters, we conducted a sensitivity analysis varying α (reconstruction weight), β (exploration/novelty weight), γ (safety penalty weight), and δ (metabolic cost

weight) across five values each, running 30 interaction rounds per configuration (600 total simulated rounds).

Results. Figure 9 summarizes the findings. Diversity is most sensitive to α (range: 0.247–0.314) and β (0.217–0.315), confirming that the reconstruction-exploration trade-off governs the system’s ability to differentiate cognitive perspectives. Evolution rate (new agents born) is most sensitive to β : at $\beta = 0$, no new agents emerge; at $\beta \geq 1.0$, up to 3 new agents are born over 30 rounds. γ (safety) and δ (metabolic cost) have relatively modest effects on diversity within the tested ranges, suggesting that the system is robust to moderate variations in these parameters. We recommend $\beta \in [0.3, 0.5]$ as a practical range balancing exploration-driven evolution with stable multi-perspective behavior.

5. Discussion

5.1. The Single-Equation Principle

A central finding is that six traditionally independent modules are unified by Equation 4 with appropriate boundary conditions (Proposition 1). This suggests a design principle that may generalize beyond memory systems: *specify the fitness landscape, not the behavioral modules*. Complex adaptive behavior emerges when agents optimize their fitness in a well-designed environment, without requiring explicit programming of each behavior.

5.2. Limitations and Future Work

1. **Basis function coverage.** The system is bounded by its basis library. Phenomena outside this scope are distorted or ignored. Future work should investigate automated basis function proposal from large-scale conversational data.
2. **Lossy compression.** Spectral encoding is inherently lossy. Surface details (“the smell of rain that afternoon”) are irrecoverable. Hybrid architectures combining spectrum for structure with raw text for detail warrant study.

3. **Neural bridge granularity.** Token-level logit bias provides coarse modulation compared to activation steering at hidden-state level. Direct model access would enable finer-grained neural-symbolic alignment.
4. **Single-user evaluation.** Current experiments use a single user profile. Multi-user studies with distinct preference profiles are needed to validate the personalization claims.
5. **Cultural bias in basis functions.** The ten initial basis functions were defined in a Chinese/English bilingual context. Cross-cultural validation with diverse basis sets is required.

5.3. Reproducibility

All experiments were conducted on a single machine with DeepSeek-Chat API access. The complete source code, data files, experiment scripts, and figure-generation code are publicly available at <https://github.com/pzhwnm/m-engine> (to be made public upon acceptance). All experiments can be reproduced by running `python -m pytest m_engine/tests/ -v` (41 automated tests) followed by `python m_engine/tests/test_deepseek.py` (full capability verification). Hyperparameters: $d_B = 10$, $d_{\text{emb}} = 256$, $\eta = 0.1$, $\alpha = 1.0$, $\beta = 0.3$, $\gamma = 0.5$, $\delta = 0.1$, $\theta_{\text{death}} = -5.0$, $\theta_{\text{merge}} = 0.85$, $\theta_{\text{split}} = 0.05$, $\theta_{\text{birth}} = 0.3$, meso trigger interval = 10 interactions, macro check interval = 50 interactions.

5.4. Broader Impact

Transparency and accountability. ACME’s full auditability—every cognitive dimension’s activation is inspectable via CLI—represents a significant step toward explainable AI memory. Users and auditors can verify not only *what* the system remembers, but *how* it interprets those memories across cognitive dimensions. This contrasts favorably with black-box LLM memory, where facts are distributed across billions of opaque weights.

Dual-use concerns. The same transparency that enables auditing also exposes the system’s “worldview” (its basis function set) to deliberate manipulation. An adversary who understands the basis functions could craft inputs

specifically designed to activate or suppress targeted dimensions. The safety-as-fitness mechanism provides partial mitigation by penalizing abnormal activation patterns, but adversarial robustness has not been systematically evaluated and warrants dedicated study.

Environmental impact. ACME’s computational footprint is dominated by LLM API calls (one per query). For a typical interaction session of 20 queries, this amounts to approximately 20 API calls, comparable to a standard chatbot session. The embedded game-dynamics computation is lightweight ($< 1\text{ms}$ per update on commodity hardware). At scale, caching frequently-used spectra and batching embedding computations could further reduce overhead.

Accessibility. The current implementation requires LLM API access (e.g., DeepSeek or OpenAI), which introduces both cost and dependency on external services. The architecture can, however, operate with any OpenAI-compatible endpoint, including locally-hosted models via Ollama, reducing barriers to entry for researchers with limited resources.

Deployment stance. We advocate deployment in contexts where explainability is paramount (clinical decision support, educational tools, personal knowledge management) and where human oversight of the basis function library is feasible. The architecture should not be deployed as an autonomous decision-making system without additional safeguards, particularly the safety-as-fitness mechanism has only been validated in single-user, low-interaction-count settings.

6. Conclusion

We have presented ACME, a cognitive memory architecture that unifies storage, multi-perspective retrieval, preference modulation, and knowledge evolution through a single replicator dynamics equation. The system treats facts as spectral projections over interpretable basis functions and treats those basis functions as strategic agents in an evolutionary game. Empirical evaluation across eleven capability dimensions, including ablation studies, demonstrates that all claimed behaviors are verifiable and that each architectural component

contributes measurably to system performance. The work contributes both a practical system for explainable memory-augmented AI and a theoretical argument: complex cognitive behavior can emerge from a well-designed fitness landscape rather than requiring independently engineered modules.

Acknowledgments

We thank DeepSeek for API access with logprobs support, which enabled the neural-symbolic bridge experiments. The complete source code, experiment scripts, and figure-generation code are publicly available at <https://github.com/pzhwnm/m-engine> (to be made public upon acceptance).

References

- [1] J. R. Anderson, D. Bothell, M. D. Byrne, S. Douglass, C. Lebiere, and Y. Qin. An integrated theory of the mind. *Psychological Review*, 111(4):1036–1060, 2004.
- [2] Y. Bai et al. Constitutional AI: Harmlessness from AI feedback. *arXiv:2212.08073*, 2022.
- [3] C. Fernando, R. Goldstein, and E. Szathmary. The neuronal replicator hypothesis. *Neural Computation*, 22(11):2809–2857, 2010.
- [4] geometer-jones. Memory engines: The geometry of automatic abstraction. GitHub Gist, 2025. <https://gist.github.com/geometer-jones/1f121fa3dc51702347858a377e388f64>
- [5] A. Snigirov. Principia Cognitia: An axiomatic framework for the new science of mind. PhilArchive, 2025. <https://philarchive.org/rec/SNIPCA>
- [6] A. Snigirov. PC-WAVES: Fourier duality of memory and prediction. PhilArchive, 2025. <https://philarchive.org/rec/SNIPFD>

- [7] S. Borgeaud, A. Mensch, J. Hoffmann, T. Cai, E. Rutherford, K. Millican, et al. Improving language models by retrieving from trillions of tokens. *ICML*, 2022.
- [8] E. M. Bender et al. On the dangers of stochastic parrots. *FAccT*, 2021.
- [9] R. Bommasani et al. On the opportunities and risks of foundation models. *arXiv:2108.07258*, 2021.
- [10] A. d’Avila Garcez and L. C. Lamb. Neurosymbolic AI: The 3rd wave. *Artificial Intelligence*, 2020.
- [11] R. W. Gayler. Vector symbolic architectures answer Jackendoff’s challenges. *ICCS*, 2003.
- [12] A. Graves et al. Neural Turing machines. *arXiv:1410.5401*, 2014.
- [13] A. Graves et al. Hybrid computing using a neural network with dynamic external memory. *Nature*, 538(7626):471–476, 2016.
- [14] J. Hofbauer and K. Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge, 1998.
- [15] M. Jaderberg et al. Population based training of neural networks. *arXiv:1711.09846*, 2017.
- [16] P. Kanerva. Hyperdimensional computing. *Cognitive Computation*, 1(2):139–159, 2009.
- [17] B. J. Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su. HippoRAG: Neurobiologically inspired long-term memory for large language models. *NeurIPS*, 2024.
- [18] V. Karpukhin et al. Dense passage retrieval for open-domain QA. *EMNLP*, 2020.
- [19] B. Kim et al. Interpretability beyond feature attribution: TCAV. *ICML*, 2018.

- [20] P. W. Koh et al. Concept bottleneck models. *ICML*, 2020.
- [21] J. E. Laird. *The Soar Cognitive Architecture*. MIT Press, 2012.
- [22] P. Lewis et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS*, 2020.
- [23] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav. Mem0: Building production-grade AI agents with scalable long-term memory. *arXiv:2504.19413*, 2025.
- [24] L. Ouyang et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.
- [25] C. Packer et al. MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*, 2023.
- [26] J. K. Pugh et al. Quality diversity: A new frontier for evolutionary computation. *Frontiers in Robotics and AI*, 2016.
- [27] K. O. Stanley and R. Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2):99–127, 2002.
- [28] S. Sukhbaatar et al. End-to-end memory networks. *NeurIPS*, 2015.
- [29] P. D. Taylor and L. B. Jonker. Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1-2):145–156, 1978.
- [30] J. Weston et al. Memory networks. *ICLR*, 2015.
- [31] M. Yuksekgonul, M. Wang, and J. Zou. Post-hoc concept bottleneck models. *ICLR (Spotlight)*, 2023.
- [32] A. Zou et al. Universal and transferable adversarial attacks on aligned language models. *arXiv:2307.15043*, 2023.

Algorithm 1 ReplicatorDynamicsUpdate (Single Cognitive Cycle)

Require: Facts F , query embedding $\mathbf{q}$, user gain $\mathbf{u}$, model gain $\mathbf{m}$

Ensure: Updated fact spectra, updated population state

```
1:  $\tilde{\mathbf{q}} \leftarrow \mathbf{P}\mathbf{q}$  ▷ Project to basis space
2: for each basis function  $b_i \in \mathcal{B}$  do
3:    $a_i \leftarrow \text{softmax}_i(\langle \mathbf{s}_i, \tilde{\mathbf{q}} \rangle \cdot u_i \cdot m_i)$ 
4:    $N_i \leftarrow 1/(\kappa_i + 1)$  ▷ Novelty bonus
5:    $\pi_i \leftarrow \alpha a_i + \beta N_i - \gamma R_i - \delta \|\mathbf{s}_i\|_2/d_B$ 
6: end for
7:  $\bar{\pi} \leftarrow \frac{1}{|\mathcal{B}|} \sum_i \pi_i$ 
8: for each fact  $f \in F$  do
9:   for each  $b_i \in \mathcal{B}$  do
10:     $s_i^{(f)} \leftarrow \text{clip}_{[0,1]}(s_i^{(f)} + \eta \cdot a_i \cdot (\pi_i - \bar{\pi}))$ 
11:   end for
12: end for
13: if meso_trigger_condition then
14:   Cull low-energy agents; crossover high-fitness pairs; mutate
15: end if
16: if macro_check_condition then
17:   Evaluate safety risks; update  $R_i$  for all agents
18: end if
19: return Updated spectra, attention vector  $\mathbf{a}$ 
```

Table 1: Quantitative results across all eleven capabilities.

Cap.	Metric	Result	Baseline/Note
C1	Attention distribution difference (causal vs. emotional)	Top dims differ: Motivation (0.275) vs. Emotion (0.378)	Random baseline: uniform (0.2 each)
C2	Preference shift magnitude	Emotion rank: 3rd $\rightarrow$ 1st	Without gain: rank unchanged
C3	Spectrum change after 1 feedback	5/5 dims changed, mean $ \Delta = 0.0029$	Without feedback: 0 changes
C4	Explainable trace availability	10 dims $\times$ per-fact spectrum	Black-box LLM: 0 explainable dims
C5	Storage compression ratio	10 + 256 floats per fact ($\approx$ 3KB)	Raw text: 100–1000 $\times$ larger
C6	Spontaneous population evolution	1 new basis born (crossover), gen 1	Static systems: 0 evolution events
C7	Cognitive gaps detected	1 gap found (gap score 0.863)	Without novelty term: 0 gaps
C8	Safety alerts (12 checks)	0 false positives	Without detectors: no monitoring
C9	Cross-memory analogy similarity	$\cos = 0.903$ (exam-cry vs. breakup-despair)	Random pair baseline: $\cos < 0.5$
C10	Knowledge exchange round-trip	100% spectrum preservation	N/A (no comparable baseline)
C11	Neural-symbolic alignment delta	+0.35 (causality), mean +0.30	Expected under perfect alignment: 0

Table 2: Ablation study: contribution of each architectural component. “Diversity” measures the pairwise cosine distance between attention distributions of causal vs. emotional queries (higher = more distinct perspectives). “Evolution” counts spontaneous basis population events. “Safety” is the number of false alerts over 50 checks.

Configuration	Diversity $\uparrow$	Evolution $\uparrow$	Safety $\downarrow$
Full ACME	0.247	1 event	0 alerts
— Game Dynamics (static spectra)	0.091	0	0
— Safety Modulator	0.244	1 event	N/A (no monitoring)
— Neural Bridge	0.247	1 event	0
— Novelty Term ($\beta = 0$)	0.239	0	0
— Population Evolution	0.241	0	0
RAG Baseline (same LLM)	0.042	0	0

Table 3: Comparison of ACME against three baselines on perspective diversity, answer distinctness (cosine distance between answers to causal vs. emotional queries), storage cost, and query latency.

System	Diversity $\uparrow$	Distinctness $\uparrow$	Storage $\downarrow$	Latency
ACME (ours)	0.059	1.005	5 floats	15.8s
RAG	0.042	0.975	185 chars	2.9s
Pure LLM	0.000	0.986	0 (no memory)	5.9s
Mem0-like	0.000	1.043	154 chars	4.5s